OEM Parts Fulfilment Research Package (HCME)

2026-09-18 · @Someone

Purpose & scope

This package is Phase 0 research for the HCME (Heavy Construction & Mining Equipment) parts-and-service platform — no code changes in this phase. Goal: understand how the real-world OEM parts fulfilment ecosystem actually works so HCME's data model, integrations, and workflow assumptions are grounded in reality rather than guesswork.

OEMs covered: Caterpillar, Komatsu, Hitachi Construction Machinery, Volvo Construction Equipment, JCB, John Deere (construction & forestry), CNH Industrial (Case CE, New Holland Construction).

Topics researched per OEM: dealer network structure, parts fulfilment process, inventory/availability model, pricing, ordering, allocation, shipment & tracking, real public APIs/developer programs, and telematics-to-parts integration.

Confidence convention used throughout this doc:

- **Confirmed** — directly stated by an official OEM page, developer portal, or other primary source.
- **Reported** — stated by secondary sources (trade press, dealer forums, third-party integrators) that are plausible but not independently verified.
- **Unknown** — actively searched for and not found; flagged rather than guessed at.

Research was conducted via web search and direct source fetches; every factual claim below is expected to carry an inline source link. Findings are still being compiled — sections below fill in as each OEM/topic cluster completes.

OEM Profiles

Caterpillar

Independent, exclusive-territory dealers are the retail parts front line; Cat corporate runs the catalog, distribution network, and digital ordering rails behind them. **Confirmed**: Cat Dealer Network.

Fulfilment channels

- Parts.Cat.com — consumer/small-fleet web store, 24/7, 1.4M+ part numbers, real-time availability & pricing, pickup at store/24-7 locker or paid rush delivery.
- SIS 2.0 — dealer-only electronic parts/service catalog (2M+ illustrations, 1.5M part numbers), cart carries into Parts.Cat.com checkout.
- Cat Integrated Procurement (Cat IP) — enterprise punchout/EDI channel: XML-over-HTTPS hosted catalog embedded in the customer's own procurement system, auto-routes orders to the local dealer, confirms via ASN + e-invoice.

Distribution: Cat Logistics runs a global Parts Distribution Center (PDC) network (108+ facilities, 23 countries) — Morton IL, Waco TX, Spokane WA, Wamego KS, York PA, San Luis Potosí (MX), Piracicaba (BR), Dubai, Den Bosch, Grimbergen, Grenoble, Johannesburg, Moscow, Bangalore, Tianjin, Melbourne/Queensland. PDC network PDF.

Pricing: dealer-set, login-tiered (implies negotiated/volume pricing) — no public dealer-net schedule for parts. A public reference point exists via government cooperative contracts (Sourcewell/NASPO): 18–34% off list on machines; parts/service discounts are dealer- and volume-specific. Sourcewell Cat pricing PDF.

Allocation / availability guarantee: Under Customer Value Agreements, Cat's **Services Commitment** program (global rollout begins Q1 2026) guarantees next-day availability on 100K+ common parts and pays the customer up to $1,000 credit if it misses. Equipment World. No published formulaic VOR-style priority-allocation policy was found beyond this SLA; 2022–23 shortages (filters, electrical, undercarriage, emissions parts) were handled dealer-by-dealer, not via a public allocation rulebook. Equipment World — supply chain.

Shipment/tracking: rush/direct shipping for a fee; Cat IP customers get ASN + e-invoice tracking. No named carriers or consumer tracking UI/API surfaced.

Real public developer platform (strongest in the OEM set): Cat Digital Marketplace — API catalog, OAuth2 (client id/secret, 60-min tokens), self-service access request (~5–7 business days provisioning), requires Cat ID. Key APIs: VisionLink APIs (telematics:

location, hours, fault codes, utilization) and the open ISO 15143-3 / AEMP 2.0 mixed-fleet telematics standard. No public parts-catalog/ordering API was found — parts B2B integration runs through Cat IP (EDI/punchout), not an open API.

Telematics → parts: VisionLink surfaces fault codes, S•O•S fluid analysis, and maintenance due-dates, and links directly into Parts.Cat.com so a flagged issue can be clicked straight into a parts order. Toromont Cat — VisionLink.

Komatsu

Hybrid model: traditionally independent dealers, but Komatsu is actively converting territories to **company-owned** dealerships — 13 owned branches in the western US, plus a June 2025 acquisition of 6 former Core Machinery locations in AZ/southern CA. Komatsu Newsroom. Global ratio of owned vs. independent is not published.

Fulfilment: My Komatsu is the single portal for fleet monitoring + parts catalog + ordering (Genuine and Reman parts), with dealer-counter pickup or direct delivery. Backed by a **Master Parts Distribution Center in Ripley, TN** (584,000 sq ft, 24/7/365, 45,000 rack + 180,000 bin locations) plus reportedly ~8 regional US/Canada DCs (Peoria IL, Bridgeville PA, Las Vegas, Denver, Minneapolis, Portland, Savannah — aggregated, not individually confirmed). Komatsu — distribution center. A consignment program stages critical spares directly at customer/job sites. Joy spare parts.

Pricing, ordering detail, allocation, shipment: **largely undocumented publicly.** No list/dealer-net pricing schedule, no published minimum order rules, no formal allocation/VOR priority policy, and no named carriers or availability-guarantee-with-credit (unlike Cat's Services Commitment) were found. A non-official third-party blog (`komatsuus.com`, not Komatsu's real `komatsu.com` domain) cites a ~5-business-day lead time and ~30% rush premium — flagged as **unverified, not from Komatsu**, and excluded from confirmed facts.

APIs: No general developer marketplace analogous to Cat Digital Marketplace was found. Komatsu's confirmed public API surface is the open **ISO 15143-3 / AEMP 2.0** telematics standard, bundled free into My Komatsu (credentials issued as username/password, no extra developer fee documented). Equipment World. No parts-catalog or parts-ordering API exists publicly.

Telematics → parts: **KOMTRAX**/KOMTRAX Plus (free, transferable, on-machine telematics) feeds My Komatsu alongside Komatsu CARE Program (KCP) maintenance tracking; users can look up and order parts from within the same portal. No source

confirms an automated "fault code triggers order" flow — it appears to be alert-then-human-orders, not fully automatic. Komatsu Newsroom.

Key asymmetry vs. Cat: Komatsu has no public equivalent to Cat Digital Marketplace, Cat IP (EDI/punchout), or a credit-backed parts-availability SLA — repeatedly absent across targeted searches, treated as a genuine finding, not a gap to caveat away.

Hitachi Construction Machinery

Americas dealer network went **independent** in 2021 after the Deere-Hitachi JV wound down, reaching ~90% North American coverage by mid-2025. Equipment World. Historical distribution agreements grant exclusive territory covering machines **and service parts/components**. SEC EDGAR filing.

Fulfilment: tiered depot chain — dealer stock → 4 fully-stocked regional parts depots → **Central Parts Depot (Tsukuba, Japan)**, operating 24/7/365 with painting/productization/packaging services. CPD and the Ami Parts Center are **AEO customs-certified** for faster cross-border clearance. Dealers/partners access shared **Global e-Service** (machine data cradle-to-grave) and **ConSite** (maintenance-alert apps) via a dealer-set-up account; ConSite explicitly lets dealers "prepare the right part before arriving at your job site." Ordering runs through a login-gated **Dealer Portal 2.0** plus a customer-facing Selected Parts search/export tool — no evidence of EDI/punchout ordering.

Pricing: three tiers — Genuine, Remanufactured (new-part warranty, lower cost), Selected parts — but no public price list; official guidance is "contact your nearest dealer." Americas Parts page.

Allocation/shipment: no public allocation policy or availability-guarantee-with-credit found (unlike Cat/Volvo). AEO certification is the only documented fast-shipment infrastructure; no named carriers or tracking portal surfaced.

APIs: a public **Telematics API** compliant with ISO 15143-3/AEMP 2.0 exists (10-min/60-min/24-hr data frequency, mixed-fleet capable), but auth/signup/pricing details were not retrievable (API page blocked direct fetch). Telematics API. No parts/service data API distinct from telematics was found.

Telematics → parts: most explicit OEM claim in this research set — ConSite (with oil-condition sensors + trend analysis) is directly marketed as letting dealers pre-stage the correct part before a service visit, extended to mining fleets via "ConSite Mine" (with Wenco).

Volvo Construction Equipment

Independent dealer network via regional Dealer Locator. Parts distribution is **shared Volvo Group infrastructure** (also serves Volvo Trucks/Mack/UD/Volvo Penta): a central DC in **Byhalia, MS** feeds 6 US regional depots + one each in Mexico/Canada, supplying 2,000+ dealers. Volvo Trucks PDC page. A **Reno, NV** regional DC (expanded 125% in 2021) runs an exclusive overnight **Dedicated Delivery Service (DDS)** carrier reaching western-US dealer branches by ~4am; a new **$37.7M Tacoma, WA** DC opens Q1 2027, expected to cut NA delivery times 1–2 days. Volvo Group PR.

Ordering: dealers order/track via the **VDN (Volvo Dealer Network)** portal (login-gated); customers get a "Parts Online Request" contact form, implying quote-driven, non-self-serve retail pricing. VDN. Volvo Group-level EDI exists but is documented for **inbound supplier** transactions into manufacturing, not confirmed as a dealer/customer parts-ordering channel for CE aftermarket. Volvo Group EDI.

Pricing: no public list/dealer-net schedule; quote-via-dealer model, same pattern as the other OEMs researched.

Allocation/guarantee — strongest documented SLA in the OEM set: the **24-Hour Parts Guarantee** commits Class-1 (machine-down/emergency) orders in-hand within 24 hours in the continental US/Alaska (48 hours Canada/PR/Hawaii) **or freight is free**. This Class-1 vs. standard-stock-order split is the clearest formal urgency-tiering found across all seven OEMs. 24-Hour Parts Guarantee.

APIs: a real commercial developer portal, developer.volvoce.com, offering an **AEMP 2.0 / ISO 15143-3 REST API** (XML, standardized telematics) and an **Extended Asset API** (fault codes, EV battery/charging, energy data). Access is **not self-serve** — credentials are issued only after the API is commercially ordered (via dealer/Services Marketplace, exclusive channel since Sept 2025); pricing is quote-only. Both APIs are explicitly telematics-only — **no parts/service-order data is exposed**. Get Started. Note: Volvo Trucks/Buses run a separate "Volvo Connect" developer ecosystem — not the same product line as Volvo CE's APIs.

Telematics → parts: **CareTrack** (free "Start" tier on 2026MY+ machines, paid Advanced tier) tracks location/hours/utilization/fuel/CO_2. Unlike Hitachi's ConSite, Volvo's public CareTrack messaging does **not** explicitly claim automatic parts pre-staging from alerts — a notable asymmetry to flag.

JCB

2,000+ dealer depots globally; JCB is actively recruiting via a "Sponsored Dealer" program (JCB HUB tools, SalesMaster CRM, co-op marketing funds) rather than disclosing fixed territory/capital terms publicly. Sponsored Dealer. Even JCB's own e-commerce platform explicitly fulfils through the dealer network, not around it.

Fulfilment: **JCB Parts Online** (parts.jcb.com, launched April 2026, UK-first) lets customers order 24/7 with click-and-collect at a local depot or home delivery — but "orders are fulfilled by JCB's existing dealer network." Agg-Net. JCB's own claim (via secondary coverage, not independently verified) is **95% next-day parts availability worldwide** through its "World Parts Centre" automation. Americas distribution runs through Miami, Ontario (CA), Grimsby (ON), plus a new 275,000 sq ft **Plainfield, IL** hub (3PL-operated by DSV) built near O'Hare specifically to cut delivery times. CompactEquip.

Pricing: no public list/dealer-net data; the new platform supports card or dealer-credit-account payment, implying at least a retail vs. account-holder pricing split.

Ordering/shipment: same/next-day depot collection or standard/next-day home delivery; order tracking is a stated feature. No EDI/punchout capability, minimum-order rules, or formal availability-guarantee-with-remedy (unlike Volvo/Cat) were found.

Allocation: no public policy found.

APIs: **no public developer/API program identified** for JCB equipment (a third-party API tracker found only the unrelated Japanese "JCB" payment-card brand). API Tracker. **JCB LiveLink** telematics supports AEMP/ISO 15143-3 mixed-fleet feeds and reportedly a partner "Developer Toolkit," but access is dealer-provisioned, not self-serve, and no auth model or pricing is published — a genuine gap versus Cat/Volvo/CNH.

Telematics → parts: LiveLink surfaces maintenance alerts, usage hours, fault/alert data, and (on electric machines) battery health, explicitly framed as enabling "advance scheduling of routine service." JCB LiveLink.

John Deere

Independently-owned dealer groups hold multi-state/multi-branch territories (e.g. Doggett, RDO, Papé, Ag-Pro) and stock parts locally, replenished from regional/central Deere warehouses. No public dealer-contract/territory-exclusivity terms found (proprietary, as with every OEM researched).

Fulfilment: dealers order/check stock via **JDPoint** (24/7 per a dated IBM case study), pulling from any dealer's stock or drop-shipping direct from a Distribution Center. **PartsAdvisor** (partsadvisor.deere.com) is the parts-lookup/exploded-diagram tool. Direct-to-consumer e-commerce runs through **Shop.Deere.com**. Backbone: **Milan, IL** primary NA PDC (2.8M sq ft, ~450K orders/week) feeding 20 regional depots — a Deere-published PDF confirms this network. Deere — Key Parts Distribution Centers PDF. A third-party logistics analysis (MWPVL, **not Deere itself**) claims a 48-hour/99%-fill-rate target and a "fast-movers regional, slow-movers centralized" strategy across ~800K unique parts — flagged as REPORTED, not an official Deere-published guarantee. MWPVL.

Pricing: no public dealer-net/list schedule; Shop.Deere.com shows real retail pricing for direct-consumer categories, and time-limited promos exist (e.g. 15% off select ag parts, capped at $1,500/machine) — not a standing structure.

Ordering/shipment: Shop.Deere.com — same-day cutoff 3pm ET weekdays, standard UPS Ground (1–6 business days), expedited UPS 2nd Day Air, free shipping $50+, US-street-address only. Shipping policy. Dealer-branded portals (Papé, Blanchard, Leading Edge) layer Deere's catalog + cart on top with 24/7 self-service ordering. A **John Deere Supply Network (JDSN)** EDI portal exists (jdsn.deere.com/edi) but specific transaction sets couldn't be confirmed (JS-rendered/auth-gated).

Allocation: no published formulaic policy; "Uptime Solutions" and generic "we'll expedite special orders" dealer-page language exist but no documented SLA-with-remedy (unlike Volvo/Cat) was found.

APIs — real, structured, and the most thoroughly documented Dealer Solutions layer in the set: developer.deere.com organizes into **Precision Tech** (Equipment/Setup/Insights/Work Results — telematics-and-agronomy focused, 8+11+3+4+3+2 sub-APIs), **Supply Chain APIs** (upstream *supplier-to-Deere* sourcing/logistics, not dealer/customer-facing), **Financing APIs**, and **Dealer Solutions** (Sales & Financial, Service, **Parts** [2 APIs], Warranty [15 APIs] — dealer/trusted-supplier only). The **Parts** category = **Dealer Syndication Service** (lets a dealer's own portal embed Deere's Parts Catalog + cart, checkout returns to the dealer's site — documented endpoint `GET /{dealer-account-id}/cart/{cart-id}`, OAuth scope `axiom`) + **Shop Deere DBS**. This is cart/checkout integration for approved dealers, **not** an open parts-lookup/pricing/inventory API for arbitrary third parties.

Auth: OAuth2 three-legged (12-hour access tokens, 365-day refresh tokens, `offline_access` scope) or client-credentials machine-to-machine. **Sandbox is real production data**, not mocked (capped at 1 test org/app, 5 connected orgs, 150K calls/month, 6-month max before requiring a production request). **Production access requires business validation, a Deere-reviewed test, and a signed Production API Agreement via DocuSign** — i.e., self-serve sandbox but a genuine partnership gate for production. No public price list; portal references "API licensing for most APIs" case-by-case. [Get Started docs, read live via browser].

Telematics → parts: **JDLink** (free connectivity on eligible modems) + **Operations Center** (free access) surface **Expert Alerts** (predictive analytics across "thousands of connected machines") and let dealers remotely read DTCs so technicians "arrive on-site with the right parts." A Sept-2025 trade headline indicates Deere recently tightened parts-ordering directly into Operations Center ("Streamlines Parts Ordering within Operations Center") — not independently verified in depth in this pass. No public API was found that auto-triggers a parts order from a fault/DTC signal.

CNH Industrial (Case Construction / New Holland Construction)

Independent dealers; New Holland Construction's NA network spans 350 dealer groups / ~725 locations, targeting 1,000 locations within five years. Equipment World. CNH is currently "purpling" — unifying Case IH/New Holland commercial management while keeping brands distinct at retail. Dealers explicitly **retain responsibility for order fulfillment** even on CNH's own e-commerce platform. CNH Newsroom.

Fulfilment: **Case Parts Store Online** (US) needs no business account, just a card + shipping address; ships from a Midwest hub near Case's warehouses, standard delivery "a few days," special routing available for emergencies. Case Parts Store Online. EMEA's **MyCNHiStore** (since 2022) supports multi-brand (Case IH/STEYR/New Holland) carts with full price/availability/shipping-cost transparency pre-purchase. Backbone: a **Lebanon, IN Master Depot** + Centralized Receiving/Packaging Center + 11 regional facilities supplying NA dealers daily (site chosen for one-day-drive proximity to 75%+ of CNH's NA supply base). For Construction Pros.

Pricing/allocation/shipment: no public list/dealer-net schedule or formal allocation policy found; CNH does describe proactively expediting **supplier→depot** shipments, but not a documented depot→dealer allocation rule. No named carriers or availability-guarantee-with-remedy found.

APIs — the most technically documented in the whole OEM set: the **CNH Developer Portal** (develop.cnh.com) offers **FieldOps API**, a unified ISO 15143-3–compliant REST API covering equipment/fleet telemetry, company hierarchy, operations data, and webhooks (superseding older separate Ag/Connect APIs). **OAuth 2.0 Authorization Code flow**: production tokens expire in 3,600s, staging in 21,600s, refresh tokens via `offline_access` scope, revocation endpoint available. Documented **rate limit: 120 concurrent users** (HTTP 429 on excess). Tokens guide, FAQ. Construction-equipment access is gated through the dealer/owner ecosystem (must be an invited Administrator on a MyCaseConstruction/MyNewHollandConstruction account tied to a dealer-established company account) — not open self-serve for arbitrary third parties, though virtual test-vehicle data is available for development. Pricing for API access is not disclosed. **FieldOps does not expose a parts-catalog or parts-ordering endpoint** — scope is telemetry/fleet/agronomic data only; parts ordering stays on the separate MyCNHiStore/Case Parts Store Online commerce layer.

Telematics → parts: **MyNewHollandConstruction** (successor to FleetForce, standard from MY2026, all MY2026 machines ship telematics-ready with a 7-year connectivity subscription) is explicitly framed to ensure "the right parts and support are available when needed." Equipment World.

Cross-check vs. JCB: CNH is the inverse of JCB — CNH has the strongest documented developer API (FieldOps) but no confirmed public parts-commerce API; JCB has a strong customer-facing parts e-commerce layer but essentially no public developer/API surface.

Ecosystem

Logistics / ERP / WMS / integration ecosystem

EDI standards: ASC X12 (ANSI-accredited, 300+ transaction sets) is the North American backbone. X12 Transaction Sets. Core parts-relevant sets: **850** PO, **855** PO Ack, **856** ASN (Advance Ship Notice), **810** Invoice, **846** Inventory Inquiry/Advice, **852** Product Activity Data, **997** Functional Ack — typical flow 850→997→855→856→810. TrueCommerce. International equivalents run on **UN/EDIFACT**: ORDERS↔850, DESADV↔856, INVOIC↔810. Astera.

Punchout/cXML: buyer's e-procurement system authenticates into the supplier's catalog via `PunchOutSetupRequest`, shops live, and gets a `PunchOutOrderMessage` cart back. PunchOutCommerce. SAP-originated **OCI (Open Catalog Interface)** is a common alternative. **Confirmed real-world HCME example**: Caterpillar's **Cat IP** explicitly

supports both **cXML PunchOut** (Ariba/Oracle) and **OCI RoundTrip** (SAP), auto-routing orders to the local dealer with ASN + e-invoice return. Cat IP. No equivalent punchout capability was confirmed publicly for Komatsu, Volvo CE, Deere, JCB, Hitachi, or CNH parts specifically in this research pass.

ERP platforms: **SAP** — DBM (Dealer Business Management, part of SAP for Automotive) covers dealer sales/service/parts/CRM; SAP Service Parts Planning (SPP) handles spare-parts network replenishment. SAP DBM Master Guide. **Infor CloudSuite Equipment** is a purpose-built multi-tenant cloud ERP for equipment OEMs/dealers/rental/distribution, covering parts pricing/quoting/sales/rental/service. Infor CloudSuite Equipment. Heavy-equipment-specific DMS vendors: **Karmak (Fusion)**, **CDK Global (Heavy Truck DMS)**, **VitalEdge IntelliDealer** (2023 merger of e-Emphasys + IntelliDealer, purpose-built for ag/construction/forestry/mining dealers), **Flyntlok**. "PSC/DIS/Every/Titan" from the original brief could not be independently confirmed as distinct current products — flagged for follow-up.

WMS patterns: wave picking, cross-docking (routing fast-moving parts straight to outbound without put-away), and cycle counting with ABC classification are standard practice; no named commercial WMS product (Manhattan, Blue Yonder, Körber) was confirmed in use at any specific OEM's PDC — Caterpillar is reported (secondary source) to run a **proprietary** inventory/order-routing system across its PDC network. IndustryNet.

Telematics data standard: **AEMP 2.0**, formally adopted as **ISO 15143-3**, defines a common JSON/XML schema (~20 parameters: location, hours, fuel, faults, idle time, payload, engine temp) over a standard web-service interface so mixed-fleet software can consume multiple OEM feeds without per-brand connectors. AEM — ISO 15143 page. **Confirmed OEM adoption**: Caterpillar (VisionLink), John Deere (own developer portal), CNH (FieldOps). Reported (not independently primary-verified) for Volvo, Komatsu, Hitachi via VisionLink's mixed-fleet integration options.

Distribution topology: consistent central-DC → regional-DC → dealer-warehouse → customer pattern, confirmed with named real facilities — Cat (Morton IL hub; regional PDCs at York PA, Spokane WA, Waco TX), Deere (Milan IL primary NA PDC, 2.8M sq ft, ~450K orders/week, feeding 20 regional depots), Volvo (Byhalia MS → Reno/Tacoma regional), CNH (Lebanon IN master depot → 11 regional facilities), JCB (Plainfield IL + Miami/Ontario CA/Grimsby ON), Hitachi (Tsukuba CPD → 4 regional depots). Replenishment logic follows standard **DRP (Distribution Requirements Planning)** —

time-phased net-requirements calculation per network node — typically blended with **min/max reorder-point** rules given HCME's lumpy, downtime-cost-driven demand pattern. NetSuite — DRP.

Aftermarket catalog standards (adjacent, not HCME-native): **PIES** (Product Information Exchange Standard, product/spec data) and **ACES** (Aftermarket Catalog Exchange Standard, vehicle fitment/application XML), governed by the Auto Care Association, plus **TecDoc** (TecAlliance, Europe-originated multi-brand catalog, 100K+ vehicle types). These are automotive-aftermarket standards — no heavy-equipment-specific equivalent standards body was found — but the underlying pattern (separate product-attribute standard + separate fitment/compatibility standard) is directly analogous to what an HCME parts catalog needs (part ↔ machine serial/model/configuration fitment). AutoCare.org — PIES, ACES, TecAlliance — TecDoc.

EDI VAN / B2B middleware vendors: SPS Commerce, TrueCommerce, OpenText Trading Grid (largest legacy VAN, enterprise-scale, MFT + EDI + API bridging) are the major named platforms bridging OEM and independent dealer/customer systems where no direct EDI/punchout integration exists. Cleo — Best EDI Software 2026.

Flagged gaps: "iDEAL" middleware (named in the original brief) could not be verified as a real heavy-equipment-parts platform under that name. EDIFACT counterparts to X12 846/852 were not independently confirmed.

Cross-OEM comparison

OEM	Dealer model	Customer-facing parts portal	Public developer API	Auth model	Formal availability SLA / guarantee	Parts data in API?
Caterpillar	Independent, exclusive territory	Parts.Cat.com (real-time stock+price) + Cat IP (cXML/OCI punchout)	Cat Digital Marketplace — VisionLink API, ISO 15143-3	OAuth2, self-service request (~5–7 days)	**Yes** — CVA Services Commitment, next-day parts, $1,000 credit on miss	No (telematics only)
Komatsu	Hybrid: independent + growing company-owned	My Komatsu	ISO 15143-3 only, bundled free	Username/password credentials	Not found	No
Hitachi CM	Independent (Americas, since 2021)	Dealer Portal 2.0 + Selected Parts search (no self-checkout confirmed)	Telematics API (ISO 15143-3)	Unknown (page blocked)	Not found	No

OEM	Dealer model	Customer-facing parts portal	Public developer API	Auth model	Formal availability SLA / guarantee	Parts data in API?
Volvo CE	Independent	VDN (dealer-only) + "Parts Online Request" form (quote-driven)	developer.volvoce.com — AEMP 2.0 + Extended Asset API	Commercial order → credentials issued (not self-serve)	**Yes** — 24-Hour Parts Guarantee, freight-free if missed	No (telematics only)
JCB	Independent, active recruitment	JCB Parts Online (launched 2026, dealer-fulfilled)	**None found**	N/A	Not found (95% next-day is a reported OEM claim, unverified)	N/A
John Deere	Independent, territory-based	Shop.Deere.com + dealer-branded portals (Papé, etc.)	developer.deere.com — Precision Tech + **Dealer Solutions Parts API** (2 APIs)	OAuth2 (12hr token/365-day refresh) + client-credentials; **production requires signed agreement**	Not found officially (48hr/99% fill is third-party MWPVL analysis, not Deere-published)	**Partially** — Dealer Syndication Service is cart/checkout integration, gated to approved dealers/suppliers
CNH (Case CE / New Holland)	Independent, undergoing brand-network unification	Case Parts Store Online / MyCNHiStore (real-time price+availability)	develop.cnh.com — FieldOps API (ISO 15143-3)	OAuth2 auth-code, documented token TTLs, 120-concurrent-user rate limit	Not found	No (FieldOps is telemetry-only; parts stay on separate commerce layer)

Pattern across all seven: dealers are universally the fulfilment layer — even OEM-run e-commerce (Parts.Cat.com, Shop.Deere.com, Case Parts Store Online, JCB Parts Online) routes physical fulfilment through the local dealer. No OEM publishes a general-purpose, self-serve **parts** API (catalog, live pricing, or ordering) open to arbitrary third parties — every public developer program found is telematics/equipment-data only (ISO 15143-3/AEMP 2.0 is the one true cross-OEM common ground). Where a "parts API" exists at all (Deere's Dealer Syndication Service), it's a cart/checkout embed for approved dealers, not a catalog/inventory feed. Only Cat (Cat IP) has confirmed EDI/punchout for enterprise procurement; Cat and Volvo are the only two with a published, credit-backed availability SLA.

Assessment

Confirmed vs. Unknown ledger

Solidly confirmed (multi-source or primary-page verified):

- All seven OEMs run through independent (or increasingly hybrid/semi-company-owned, e.g. Komatsu) dealer networks that remain the physical fulfilment layer even when the OEM runs its own e-commerce front end.

- ISO 15143-3 / AEMP 2.0 is the real, working cross-OEM telematics data standard — confirmed adopted by Cat, Deere, CNH directly, and available to Komatsu/Volvo/Hitachi/JCB via mixed-fleet platforms.
- Caterpillar (Cat Digital Marketplace, Cat IP), Volvo CE (developer.volvoce.com), John Deere (developer.deere.com), and CNH (develop.cnh.com FieldOps) each have real, documented public developer portals with OAuth2 and defined onboarding paths.
- No OEM publishes an open, self-serve parts catalog/pricing/inventory API for third parties. Every public API found is either telematics-only or (Deere) a gated dealer cart-embed.
- Cat's CVA Services Commitment and Volvo's 24-Hour Parts Guarantee are the only two confirmed, published, credit/remedy-backed parts-availability SLAs across all seven OEMs.
- Named real distribution facilities exist and were verified for Cat, Deere, Volvo, CNH, JCB, Hitachi, Komatsu (see Ecosystem section).

Genuinely unknown (searched, not found — treat as real gaps, not oversights):

- Public list/dealer-net/customer pricing schedules for parts, for **any** of the seven OEMs. All pricing is dealer-quoted or login-gated.
- A formal, published cross-dealer parts-allocation policy (VOR-style priority rules) for any OEM except the implicit Class-1 tiering in Volvo's guarantee terms.
- Named shipping carriers used by OEM parts networks (Cat, Komatsu, Hitachi, JCB, CNH, Deere all undocumented publicly; Volvo names an exclusive DDS carrier for one region only).
- Whether any OEM's telematics fault-code alert **automatically** (i.e., without a human in the loop) triggers a parts order — every OEM markets "alert → dealer prepares/orders part," none confirms full automation.
- JCB's and Komatsu's API auth models/pricing in detail (pages blocked or simply not published).
- Exact EDI transaction-set usage for Deere's JDSN and Volvo Group EDI as applied specifically to aftermarket parts (vs. inbound supplier logistics).
- Whether "iDEAL" middleware and the DMS names "PSC/DIS/Every/Titan" (from the original research brief) refer to real, currently-active products — not independently verified; worth confirming the intended vendors before citing further.

Source-quality flags (do not treat as OEM-confirmed facts): the `komatsuus.com` "rush order" blog is not Komatsu's official domain; Warren CAT's 97%-in-24-hours and 26M-part figures describe one dealer, not a company-wide Cat metric; MWPVL's 99%/48-hour

Deere fill-rate claim is third-party logistics analysis, not Deere-published.

Implications for HCME design

1. **Design around the dealer, not around the OEM.** Every OEM researched fulfils through dealers, even on their own e-commerce platforms. HCME's data model should treat "dealer" as a first-class fulfilment entity (territory, stock, pricing authority) rather than a thin pass-through to an OEM warehouse.
2. **Don't assume a parts API will ever exist.** No OEM exposes open parts catalog/pricing/inventory data programmatically. Any HCME integration with real OEM data will need to be either (a) a negotiated EDI/punchout relationship (Cat IP is the only confirmed precedent, via cXML or OCI), (b) a dealer-portal-mediated manual/semi-automated workflow, or (c) built on top of the one thing that *is* open — ISO 15143-3/AEMP 2.0 telematics data — to infer maintenance/parts need rather than to query parts stock directly.
3. **Telematics-to-parts is the one automatable link that's real across OEMs.** All seven either confirm or plausibly support fault-code/maintenance-alert data via ISO 15143-3. HCME's most realistic "proactive parts" feature is condition-based alerting that prompts a human-initiated order, not a fully automatic OEM-side purchase — no OEM was confirmed to do the latter.
4. **Pricing must be modeled as dealer-quoted, not centrally published.** No OEM publishes list/dealer-net pricing; HCME should treat price as a per-dealer, per-account negotiated value, not a global catalog field synced from an OEM feed.
5. **Availability SLAs are the exception, not the rule.** Only Cat and Volvo have real, credit-backed guarantees; treat any "X-hour parts guarantee" concept in HCME as inspired by these two, not as an industry universal.
6. **EDI (ASC X12: 850/855/856/810/846/852) is the realistic large-account integration path**, not REST APIs, for any B2B parts-ordering integration with dealers/distributors that already run a DMS (Karmak, VitalEdge IntelliDealer, Infor CloudSuite Equipment, SAP DBM).

7. **Suggested Phase 1 next steps**: (a) directly contact 2–3 dealers (any brand) to verify what their DMS/parts-ordering stack actually looks like day-to-day — this research is necessarily outside-in; (b) evaluate whether Cat IP or CNH FieldOps is worth a real partner-access request, since they're the two most concretely documented integration points; (c) decide whether HCME's own parts-availability commitment (if any) should mirror the Cat/Volvo SLA pattern; (d) clarify the "iDEAL"/DMS-name gaps flagged above before they're referenced in downstream design docs.

Sources

All links accessed 2026-09-18 via web search and direct fetch. Grouped by OEM/topic; ⚠️ flags a source of lower reliability noted in the text above.

Caterpillar: Dealer Network · Parts.Cat.com · SIS 2.0 · Cat IP · Cat IP White Paper PDF · Global PDC network PDF · Sourcewell pricing PDF · Services Commitment · Supply chain shortages · Cat Digital Marketplace · VisionLink API guide · ISO 15143-3 FAQ · VisionLink product page · ⚠️ Warren CAT dealer figures

Komatsu: My Komatsu · Komatsu Parts · Company-owned Dealer West · Core Machinery acquisition · Distribution center · Joy spare parts / consignment · KOMTRAX · My Komatsu telematics · Equipment World — mixed-fleet telematics · ⚠️ komatsuus.com rush-order blog (not official domain)

Hitachi Construction Machinery: Global Network · Parts & Service · Equipment World — dealer network · ConSite Americas · Global e-Service · Selected Parts tool · Americas Parts page · Telematics API · ConSite Mine / Wenco

Volvo CE: Dealer Locator · Volvo Trucks PDC page (shared network) · Reno DC expansion · Tacoma DC announcement · VDN portal · 24-Hour Parts Guarantee · Volvo Group EDI · Developer Portal — API Products · Get Started · CareTrack

JCB: Sponsored Dealer · JCB Parts Online launch — Agg-Net · JCB Parts Online — Landscape & Amenity · Plainfield IL distribution hub · JCB LiveLink · API Tracker (no JCB equipment API found)

John Deere: Key Parts Distribution Centers PDF · Shop.Deere.com shipping policy · PartsAdvisor · developer.deere.com · ⚠️ MWPVL — Deere distribution strategy (third-party analysis) · Construction Equipment magazine — Operations Center · Logistics Viewpoints — Deere parts/dealer coordination

CNH Industrial: Case Parts Store Online · CNH Newsroom — MyCNHiStore launch · Lebanon IN Master Depot · New Holland dealer expansion · MyNewHollandConstruction telematics · CNH Developer Portal — FieldOps · FieldOps Tokens guide · CE Brand Portal getting-started

Ecosystem / logistics / standards: X12 Transaction Sets · TrueCommerce EDI codes · Astera — EDIFACT vs X12 · PunchOutCommerce — cXML · SAP DBM Master Guide PDF · Infor CloudSuite Equipment · AEM — ISO 15143-3 standard page · NetSuite — DRP · AutoCare.org — PIES · AutoCare.org — ACES · TecAlliance — TecDoc · Cleo — Best EDI Software 2026